Titanic EDA

In [82]:
```
import pandas as pd
```

In [83]:
```
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
```

In [84]:
```
titanic_data=pd.read_csv('titanic_data.csv')
```

In [85]:
```
titanic_data.describe()
```

Out[85]:

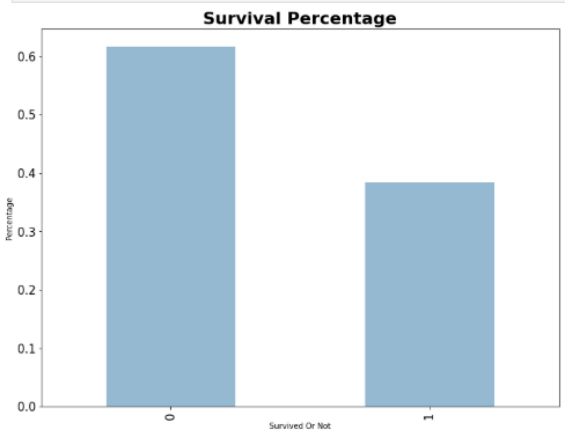
	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

In [86]:
```
titanic_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   PassengerId  891 non-null    int64  
 1   Survived     891 non-null    int64  
 2   Pclass       891 non-null    int64  
 3   Name         891 non-null    object 
 4   Sex          891 non-null    object 
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64  
 7   Parch        891 non-null    int64  
 8   Ticket       891 non-null    object 
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object 
 11  Embarked     889 non-null    object 
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

In [87]:
```
font = {
    'weight' : 'bold',
    'size'   : 22
}
```

In [88]:
```
plt.figure(figsize=(12,9))
titanic_data['Survived'].value_counts(normalize=True).plot(kind='bar',alpha=0.5)
plt.xticks(size=15)
plt.yticks(size=15)
plt.xlabel('Survived Or Not')
plt.ylabel('Percentage ')
plt.title("Survival Percentage", fontdict=font)
# plt.legend(loc='best')
plt.savefig('Survival.png')
plt.show()
```



In [89]:
```
import numpy as np
```

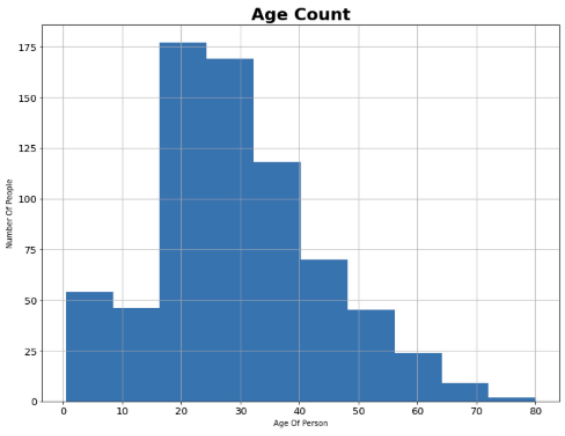
In [90]:
```
titanic_data.isnull().any()
```

Out[90]:
```
PassengerId    False
Survived       False
Pclass         False
Name           False
Sex            False
Age             True
SibSp          False
Parch          False
Ticket         False
Fare           False
Cabin           True
Embarked        True
dtype: bool
```

In [91]:
```
print("age",titanic_data.Age.isna().sum())
print("cabin",titanic_data.Cabin.isna().sum())
print("embark",titanic_data.Embarked.isna().sum())
```

```
age 177
cabin 687
embark 2
```

In [92]:
```
plt.figure(figsize=(12,9))
titanic_data.Age.hist()
plt.xticks(size=13)
plt.yticks(size=13)
plt.xlabel('Age Of Person')
plt.ylabel('Number Of People')
plt.title("Age Count", fontdict=font)
# plt.legend(loc='best')
plt.savefig('Age.png')
plt.show()
```

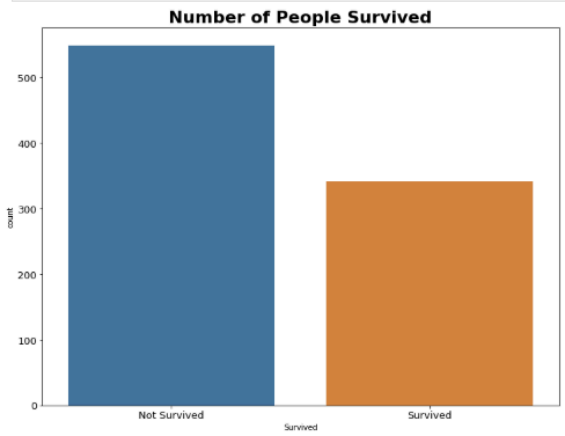


In [93]:
```
titanic_data.head()
```

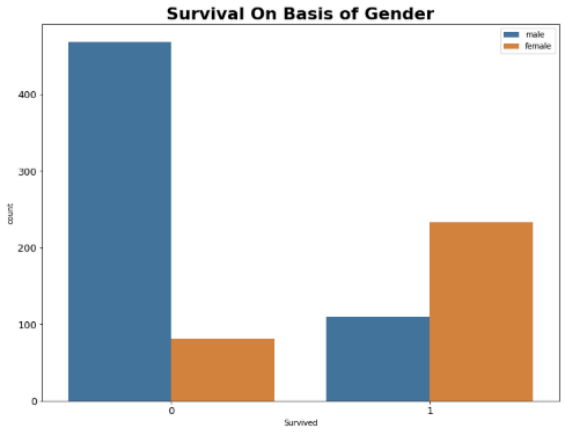
Out[93]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

In [94]:
```
plt.figure(figsize=(12,9))
sns.countplot(x='Survived',data=titanic_data)
label=['Not Survived','Survived']
plt.xticks(titanic_data['Survived'].unique(), label, size=13)
plt.yticks(size=13)
plt.title("Number of People Survived", fontdict=font)
plt.savefig('survived_people.png')
plt.show()
```

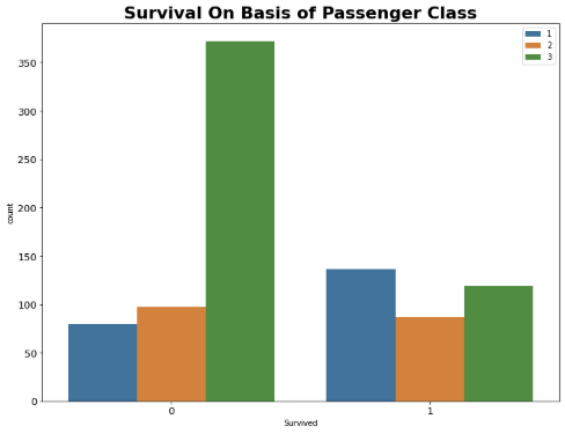


In [95]:
```
plt.figure(figsize=(12,9))
sns.countplot(x='Survived',hue='Sex',data=titanic_data)
plt.xticks(size=13)
plt.yticks(size=13)
plt.title("Survival On Basis of Gender", fontdict=font)
plt.legend(loc='best')
plt.savefig('Survival_gender.png')
plt.show()
```



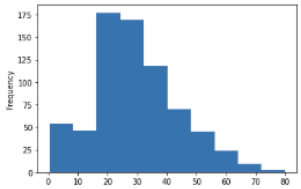
In [96]:
```
plt.figure(figsize=(12,9))

sns.countplot(x='Survived',hue='Pclass',data=titanic_data)
plt.xticks(size=13)
plt.yticks(size=13)
plt.title("Survival On Basis of Passenger Class", fontdict=font)
plt.legend(loc='best')
plt.savefig('Survival_Pclass.png')
plt.show()
```



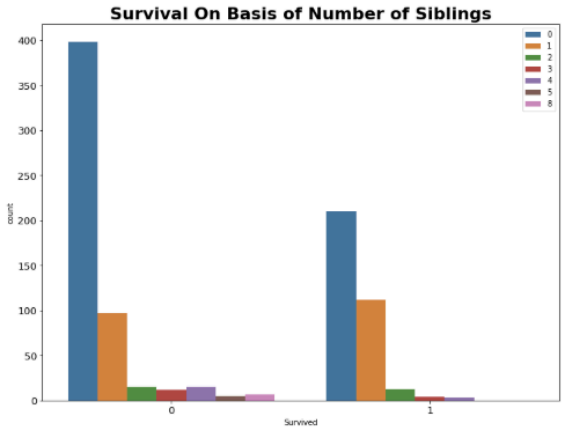
In [97]:
```
titanic_data['Age'].plot.hist()
```

Out[97]:
```
<AxesSubplot: ylabel='Frequency'>
```

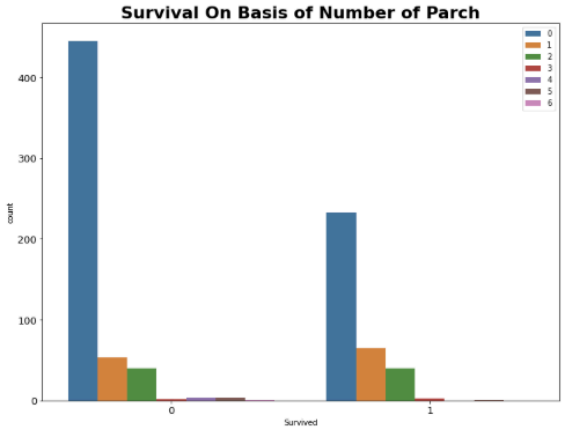


In [98]:
```
plt.figure(figsize=(12,9))


sns.countplot(x='Survived',hue='SibSp',data=titanic_data)
plt.xticks(size=13)
plt.yticks(size=13)
plt.title("Survival On Basis of Number of Siblings", fontdict=font)
plt.legend(loc='best')
plt.savefig('Survival_sibling.png')
plt.show()
```



In [99]:
```
plt.figure(figsize=(12,9))
sns.countplot(x='Survived',hue='Parch',data=titanic_data)
plt.xticks(size=13)
plt.yticks(size=13)
plt.title("Survival On Basis of Number of Parch", fontdict=font)
plt.legend(loc='best')
plt.savefig('Survival_parch.png')
plt.show()
```



Data Cleaning

In [100…
```
titanic_data.isnull().any()
```

Out[100…
```
PassengerId    False
Survived       False
Pclass         False
Name           False
Sex            False
Age             True
SibSp          False
Parch          False
Ticket         False
Fare           False
Cabin           True
Embarked        True
dtype: bool
```

In [101…
```
titanic_data.isnull().sum()
```

Data Cleaning

In [100…
```
titanic_data.isnull().any()
```

Out[100…
```
PassengerId    False
Survived       False
Pclass         False
Name           False
Sex            False
Age             True
SibSp          False
Parch          False
Ticket         False
Fare           False
Cabin           True
Embarked        True
dtype: bool
```

In [101…
```
titanic_data.isnull().sum()
```

Out[101…
```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

In [102…
```
titanic_data.drop('Cabin',axis=1,inplace=True)
```

In [103…
```
titanic_data.head()
```

Out[103…

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	S

In [104…
```
titanic_data.dropna(inplace=True)
```

In [105…
```
titanic_data.isnull().sum()
```

Out[105…
```
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Embarked       0
dtype: int64
```

In [106…
```
Sex=pd.get_dummies(titanic_data['Sex'])
Sex.head()
```

Out[106…

	female	male
0	0	1
1	1	0
2	1	0
3	1	0
4	0	1

In [107…
```
Passengerclass=pd.get_dummies(titanic_data['Pclass'])
Passengerclass.head()
```

Out[107…

	1	2	3
0	0	0	1
1	1	0	0
2	0	0	1
3	1	0	0
4	0	0	1

In [108…
```
Embark=pd.get_dummies(titanic_data['Embarked'])
Embark.head()
```

Out[108…

	C	Q	S
0	0	0	1
1	1	0	0
2	0	0	1
3	0	0	1
4	0	0	1

In [109…
```
titanic_data=pd.concat([titanic_data,Sex,Passengerclass,Embark],axis=1)
```

In [110…
```
titanic_data.head()
```

Out[110…

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Embarked	female	male	1	2	3	C	Q	S
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	S	0	1	0	0	1	0	0	1
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C	1	0	1	0	0	1	0	0
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	S	1	0	0	0	1	0	0	1
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	S	1	0	1	0	0	0	0	1
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	S	0	1	0	0	1	0	0	1

In [111…

In [111…
```
titanic_data.drop(['PassengerId','Pclass','Name','Ticket','Embarked','Sex'],axis=1,inplace=True)
```

In [112…
```
titanic_data.head()
```

Out[112…

	Survived	Age	SibSp	Parch	Fare	female	male	1	2	3	C	Q	S
0	0	22.0	1	0	7.2500	0	1	0	0	1	0	0	1
1	1	38.0	1	0	71.2833	1	0	1	0	0	1	0	0
2	1	26.0	0	0	7.9250	1	0	0	0	1	0	0	1
3	1	35.0	1	0	53.1000	1	0	1	0	0	0	0	1
4	0	35.0	0	0	8.0500	0	1	0	0	1	0	0	1

Applying Logistic Regression Model to Predict the Survival of Passenger

In [113…
```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
```

In [114…
```
X,Y=titanic_data.drop(['Survived','C','Q','S','male'],axis=1),titanic_data.Survived
```

In [115…
```
X.head()
```

Out[115…

	Age	SibSp	Parch	Fare	female	1	2	3
0	22.0	1	0	7.2500	0	0	0	1
1	38.0	1	0	71.2833	1	1	0	0
2	26.0	0	0	7.9250	1	0	0	1
3	35.0	1	0	53.1000	1	1	0	0
4	35.0	0	0	8.0500	0	0	0	1

In [116…
```
X.shape[0]
```

Out[116… 712

In [ ]:
```
```

In [117…
```
Y.shape[0]
```

Out[117… 712

In [118…
```
x_train,x_test,y_train,y_test=train_test_split(X,Y,test_size=0.2,random_state=42)
```

In [119…
```
model=LogisticRegression()
```

In [120…
```
model.fit(x_train,y_train)
```

Out[120…
```
LogisticRegression()
```
In a Jupyter environment, please rerun this cell to show the HTML representation or trust the notebook.
On GitHub, the HTML representation is unable to render, please try loading this page with nbviewer.org.

In [121…
```
prediction=model.predict(x_test)
```

In [122…
```
prediction
```

Out[122…
```
array([1, 1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 1, 0, 1, 1, 0, 1,
       1, 0, 0, 1, 0, 1, 0, 1, 1, 0, 0, 0, 0, 0, 1, 1, 0, 1, 0, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 1, 0, 0,
       0, 0, 1, 0, 1, 0, 0, 0, 0, 1, 0, 0, 1, 0, 1, 1, 0, 0, 0, 0, 1,
       0, 1, 0, 0, 0, 0, 0, 1, 1, 0, 1, 1, 1, 0, 1, 0, 0, 0, 0, 0, 0,
       0, 1, 0, 0, 1, 0, 0, 0, 1, 1, 0, 1, 1, 1, 0, 1, 0, 1, 0, 1, 1,
       0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 0])
```

In [123…
```
x_test.head(3)
```

Out[123…

	Age	SibSp	Parch	Fare	female	1	2	3
641	24.0	0	0	69.3000	1	1	0	0
496	54.0	1	0	78.2667	1	1	0	0
262	52.0	1	1	79.6500	0	1	0	0

In [124…
```
dataf=[456,24.0,1,1,67.3400,0,0,0,0,0]
```

In [125…
```
testing=pd.DataFrame(dataf)
```

In [126…
```
from sklearn.metrics import accuracy_score
```

In [127…
```
acc_logreg = round(accuracy_score(prediction, y_test) * 100, 2)
print(acc_logreg)
```

80.42

In [128…
```
test=pd.read_csv('test.csv')
test.head()
```

Out[128…

	PassengerId	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	892	3	Kelly, Mr. James	male	34.5	0	0	330911	7.8292	NaN	Q
1	893	3	Wilkes, Mrs. James (Ellen Needs)	female	47.0	1	0	363272	7.0000	NaN	S
2	894	2	Myles, Mr. Thomas Francis	male	62.0	0	0	240276	9.6875	NaN	Q
3	895	3	Wirz, Mr. Albert	male	27.0	0	0	315154	8.6625	NaN	S
4	896	3	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.0	1	1	3101298	12.2875	NaN	S

In [129…
```
X.head()
```

Out[129…

	Age	SibSp	Parch	Fare	female	1	2	3
0	22.0	1	0	7.2500	0	0	0	1
1	38.0	1	0	71.2833	1	1	0	0
2	26.0	0	0	7.9250	1	0	0	1
3	35.0	1	0	53.1000	1	1	0	0
4	35.0	0	0	8.0500	0	0	0	1

In [130…
```
clean_test = test[['PassengerId', 'Age', 'SibSp', 'Parch', 'Fare', 'Sex', 'Pclass', 'Embarked']]
clean_test.head()
```

Out[130…

	PassengerId	Age	SibSp	Parch	Fare	Sex	Pclass	Embarked
0	892	34.5	0	0	7.8292	male	3	Q
1	893	47.0	1	0	7.0000	female	3	S
2	894	62.0	0	0	9.6875	male	2	Q
3	895	27.0	0	0	8.6625	male	3	S
4	896	22.0	1	1	12.2875	female	3	S

In [131…
```
cleaning = pd.get_dummies(clean_test[['Sex', 'Pclass', 'Embarked']])
cleaning
```

Out[131…

	Pclass	Sex_female	Sex_male	Embarked_C	Embarked_Q	Embarked_S
0	3	0	1	0	1	0
1	3	1	0	0	0	1
2	2	0	1	0	1	0
3	3	0	1	0	0	1
4	3	1	0	0	0	1
...	...	...	...	...	...	...
413	3	0	1	0	0	1
414	1	1	0	1	0	0
415	3	0	1	0	0	1
416	3	0	1	0	0	1
417	3	0	1	1	0	0

418 rows × 6 columns

In [132…
```
class_dummies = pd.get_dummies(clean_test['Pclass'])
class_dummies.head()
```

Out[132…

	1	2	3
0	0	0	1
1	0	0	1
2	0	1	0
3	0	0	1
4	0	0	1

In [133…
```
dummies = pd.concat([class_dummies, cleaning], axis=1)
dummies.head()
```

Out[133…

	1	2	3	Pclass	Sex_female	Sex_male	Embarked_C	Embarked_Q	Embarked_S
0	0	0	1	3	0	1	0	1	0
1	0	0	1	3	1	0	0	0	1
2	0	1	0	2	0	1	0	1	0
3	0	0	1	3	0	1	0	0	1
4	0	0	1	3	1	0	0	0	1

In [134…
```
dummies.shape
```

Out[134… (418, 9)

In [135…
```
clean_test = pd.concat([clean_test, dummies], axis=1)
clean_test.head()
```

Out[135…

	PassengerId	Age	SibSp	Parch	Fare	Sex	Pclass	Embarked	1	2	3	Pclass	Sex_female	Sex_male	Embarked_C	Embarked_Q	Embarked_S
0	892	34.5	0	0	7.8292	male	3	Q	0	0	1	3	0	1	0	1	0
1	893	47.0	1	0	7.0000	female	3	S	0	0	1	3	1	0	0	0	1
2	894	62.0	0	0	9.6875	male	2	Q	0	1	0	2	0	1	0	1	0
3	895	27.0	0	0	8.6625	male	3	S	0	0	1	3	0	1	0	0	1
4	896	22.0	1	1	12.2875	female	3	S	0	0	1	3	1	0	0	0	1

In [136…
```
clean_test.shape
```

Out[136… (418, 17)

In [137…
```
X.head(2)
```

Out[137…

	Age	SibSp	Parch	Fare	female	1	2	3
0	22.0	1	0	7.2500	0	0	0	1
1	38.0	1	0	71.2833	1	1	0	0

In [138…
```
clean_test.drop(columns=['Sex', 'Pclass', 'Embarked', 'Pclass'], axis=1, inplace=True)
clean_test.head()
```

Out[138…

	PassengerId	Age	SibSp	Parch	Fare	1	2	3	Sex_female	Sex_male	Embarked_C	Embarked_Q	Embarked_S
0	892	34.5	0	0	7.8292	0	0	1	0	1	0	1	0
1	893	47.0	1	0	7.0000	0	0	1	1	0	0	0	1
2	894	62.0	0	0	9.6875	0	1	0	0	1	0	1	0
3	895	27.0	0	0	8.6625	0	0	1	0	1	0	0	1
4	896	22.0	1	1	12.2875	0	0	1	1	0	0	0	1

4	35.0	0	0	8.0500	0	0	0	1

In [140…
```
X.shape[1]
```

Out[140… 8

In [141…
```
clean_test.shape[1]
```

Out[141… 13

In [142…
```
X.head(2)
```

Out[142…

	Age	SibSp	Parch	Fare	female	1	2	3
0	22.0	1	0	7.2500	0	0	0	1
1	38.0	1	0	71.2833	1	1	0	0

In [143…
```
clean_test.drop(columns=['PassengerId'], axis=1, inplace=True)
clean_test.head()
```

Out[143…

	Age	SibSp	Parch	Fare	1	2	3	Sex_female	Sex_male	Embarked_C	Embarked_Q	Embarked_S
0	34.5	0	0	7.8292	0	0	1	0	1	0	1	0
1	47.0	1	0	7.0000	0	0	1	1	0	0	0	1
2	62.0	0	0	9.6875	0	1	0	0	1	0	1	0
3	27.0	0	0	8.6625	0	0	1	0	1	0	0	1
4	22.0	1	1	12.2875	0	0	1	1	0	0	0	1

In [144…
```
clean_test.isnull().any()
```

Out[144…
```
Age            True
SibSp         False
Parch         False
Fare           True
1             False
2             False
3             False
Sex_female    False
Sex_male      False
Embarked_C    False
Embarked_Q    False
Embarked_S    False
dtype: bool
```

In [145…
```
clean_test.Age = clean_test.Age.fillna(titanic_data['Age'].mean())
```

In [146…
```
clean_test.Fare = clean_test.Fare.fillna(titanic_data['Fare'].mean())
```

In [147…
```
clean_test.isnull().any()
```

Out[147…
```
Age           False
SibSp         False
Parch         False
Fare          False
1             False
2             False
3             False
Sex_female    False
Sex_male      False
Embarked_C    False
Embarked_Q    False
Embarked_S    False
dtype: bool
```

In [148…
```
clean_test.head()
```

Out[148…

	Age	SibSp	Parch	Fare	1	2	3	Sex_female	Sex_male	Embarked_C	Embarked_Q	Embarked_S
0	34.5	0	0	7.8292	0	0	1	0	1	0	1	0
1	47.0	1	0	7.0000	0	0	1	1	0	0	0	1
2	62.0	0	0	9.6875	0	1	0	0	1	0	1	0
3	27.0	0	0	8.6625	0	0	1	0	1	0	0	1
4	22.0	1	1	12.2875	0	0	1	1	0	0	0	1

In [149…
```
new_data=clean_test.drop(['Embarked_C','Embarked_Q','Embarked_S','Sex_male'],axis=1)
```

In [150…
```
new_data.head()
```

Out[150…

	Age	SibSp	Parch	Fare	1	2	3	Sex_female
0	34.5	0	0	7.8292	0	0	1	0
1	47.0	1	0	7.0000	0	0	1	1
2	62.0	0	0	9.6875	0	1	0	0
3	27.0	0	0	8.6625	0	0	1	0
4	22.0	1	1	12.2875	0	0	1	1

In [151…
```
final_prediction = model.predict(new_data)
```

In [152…
```
final_prediction
```

Out[152…
```
array([0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 0, 1, 0, 0, 0, 1, 0,
       1, 1, 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0,
       0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 1, 0, 1, 0, 0, 1, 0,
       0, 1, 1, 0, 0, 0, 0, 1, 1, 1, 0, 0, 1, 0, 0, 0, 1, 0, 1, 0, 0, 0,
       0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0,
       0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0,
       0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 1, 1, 0, 1, 0, 1, 0, 1, 0, 0, 0,
       0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0,
       0, 1, 0, 0, 1, 1, 1, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 1, 0,
       1, 0, 0, 0, 1, 0, 1, 1, 0, 1, 1, 0, 0, 0, 1, 0, 0, 1, 0, 1, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0,
       1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       1, 0, 1, 0, 0, 0, 1, 0, 1, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0,
       0, 1, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 1, 0,
       1, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 1, 1, 0, 0, 0, 1, 0, 1,
       0, 1, 0, 0, 0, 1, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 1, 0, 1, 0, 0, 1,
       1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 1, 1, 1, 0,
       0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1,
       0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0])
```